

Domain Adaptation for Cross-View Localization via Multi-Teacher Knowledge Distillation

CHENGLIZHAO CHEN, China University of Petroleum (East China), China

QIANXI YUAN, China University of Petroleum (East China), China

SHAOFENG LIANG, China University of Petroleum (East China), China

MENGKE SONG, China University of Petroleum (East China), China

XINYU LIU*, China University of Petroleum (East China), China

Cross-view fine-grained localization estimates a ground camera's pixel-level coordinates in aerial images by analyzing visual correspondences between views. Recent studies have made significant progress in this task, but when the models trained in a source area are directly applied to a new target area, their localization performance often suffers significant degradation due to domain gap between the two areas. Moreover, obtaining accurate ground truth (GT) for the target area to retrain the models is prohibitively expensive. To adapt the localization model to the target area, this paper proposes a weakly supervised learning approach based on multi-teacher knowledge distillation. This approach utilizes multiple pre-trained teacher models to make predictions for the target area and employs a learning-free cross-view instance matching and view alignment (CVMA) module to evaluate the quality of predicted coordinates from geometric, semantic, and visual perspectives. Based on the evaluation results, the best prediction is selected as pseudo-GT, and potential anomalous training samples are filtered out. The CVMA module also functions as a learning-free fine-grained localization method, achieving performance comparable to some learning-based methods. Our approach is validated on the VIGOR benchmark using three state-of-the-art models, and experimental results show that our method significantly improves the localization performance of models in the target area.

CCS Concepts: • **Computing methodologies** → **Scene understanding; Matching**; • **Computer systems organization** → Robotics.

Additional Key Words and Phrases: Cross-view Geo-localization, Multi-teacher knowledge Distillation, Instance Matching, View Alignment, Domain Adaptation

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*Corresponding author.

Authors' Contact Information: ChengliZhao Chen, cclz123@163.com, China University of Petroleum (East China), Qingdao, China; Qianxi Yuan, cclz123@163.com, China University of Petroleum (East China), Qingdao, China; Shaofeng Liang, shaofengliang1999@gmail.com, China University of Petroleum (East China), Qingdao, China; MengKe Song, songsook@163.com, China University of Petroleum (East China), Qingdao, China; Xinyu Liu, liuxy001005@163.com, China University of Petroleum (East China), Qingdao, China.

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1 Introduction

Cross-view localization aims to estimate the position of a ground camera by comparing a ground-level image with aerial image patches from a reference database. This task is essential for applications such as outdoor robotics [46] and autonomous driving [18, 31]. Previous works [4, 5, 9, 11, 15, 19, 20, 22, 23, 37, 49, 55, 56] formulated this task as an image retrieval problem, where the goal is to retrieve the corresponding aerial image patch from the reference database for a given ground query image. The center of the retrieved aerial image patch is then used as the estimated camera location. Cross-view image retrieval methods can effectively match ground images with aerial image patches. However, in practical applications, the camera location is unlikely to be exactly at the center of the aerial image patch, leading to significant estimation errors. Recent works [13, 17, 34–36, 38, 43, 53, 57, 60, 67] have further focused on fine-grained localization, aiming to estimate the ground camera’s pixel coordinates in the aerial image for a given pair of matched ground and aerial images. Cross-view camera fine-grained localization holds promise for correcting GPS signals or serving as an important supplement, especially in urban canyons where multi-path effects and signal blockages may lead to inaccurate GPS signals, with errors potentially reaching tens of meters [2, 58, 59].

Although recent methods [17, 35, 38, 43, 45, 53, 57, 60] have made significant progress in the fine-grained localization task, localization models trained in a source area often experience a significant performance drop when directly generalized to new target areas without ground truth (GT). This degradation primarily stems from the fact that existing models rely heavily on area-specific visual correspondences, which vary drastically across domains. Traditional domain adaptation methods, such as feature alignment, typically aim to align global feature distributions. However, this global alignment strategy inherently fails to preserve the fine-grained local geometric correspondences required for precise localization, rendering it unsuitable for this task. Additionally, collecting a large amount of accurate GT (i.e., true camera positions) to retrain models in the target area is often prohibitively expensive. To address these issues, [61] generates pseudo-GT for the target area using knowledge self-distillation. However, this approach has inherent limitations: since the teacher and student models share the same network architecture, the student tends to inherit the teacher’s systemic biases and incorrect predictions. This "homogeneous" distillation approach leads to severe confirmation bias, restricting the adaptive improvement of the model in the target area.

To adapt the localization model to the target area, we propose a weakly-supervised approach based on multi-teacher knowledge distillation. Localization models with different network architectures differ in feature extraction and inference processes, each having its own advantages. Using multiple models as teachers provides more diverse predictions for the target area, and selecting the most reliable predictions from these is the key challenge. Therefore, we design a learning-free cross-view instance matching and view alignment (CVMA) module to evaluate the quality of candidate predictions. Based on the evaluation results, we select the best prediction to generate pseudo-GT and filter out potential anomalous training samples to mitigate the negative impact of prediction errors and data noise on model performance. This core idea is conceptually illustrated in Figure 1.

The CVMA module evaluates the quality of candidate predictions from geometric, semantic, and visual perspectives. Its learning-free nature enables it to avoid being restricted by data from a specific area, exhibiting strong robustness across different areas. Specifically, for each candidate prediction: instance matching step pairs static reference instances (e.g., buildings, trees, roads) in ground and aerial images based on azimuthal relationships and semantic consistency. View alignment generates two sets of aligned image pairs through geometric transformations and cropping-resizing operations. When the predicted coordinates are close to the true camera position, the azimuth of the matched instances between the ground and aerial images strictly corresponds, and the two sets of image pairs exhibit high visual similarity.

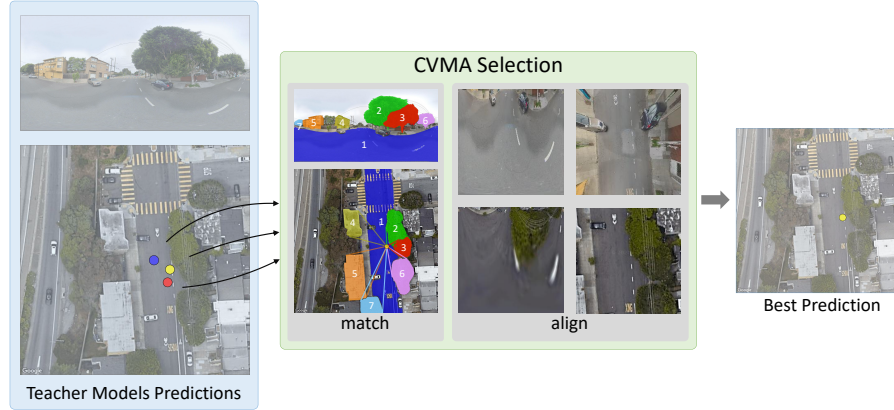


Fig. 1. CVMA performs instance matching and view alignment on candidate predicted coordinates from different teacher models. Based on the matching and alignment results, the optimal prediction is selected. When the predicted coordinates are close to the true camera position, the ground-aerial azimuths of the matched instance pairs exhibit strict correspondence, and the two sets of image pairs are visually highly similar at the given location.

Additionally, evaluating a grid-sampled set of coordinates in the aerial image with CVMA can serve as a learning-free localization method. The main contributions of this work are as follows.

- We propose a weakly supervised learning method based on multi-teacher knowledge distillation, which leverages heterogeneous teacher models to provide diverse predictions, thereby generating high-quality pseudo-GTs in the absence of target-domain annotations and mitigating the domain adaptation issue.
- We design a learning-free CVMA module that serves as the core mechanism for controlling the reliability of pseudo-GTs by strictly evaluating predictions from geometric, semantic, and visual perspectives. Additionally, the CVMA module functions as a standalone learning-free localization method, achieving accuracy comparable to some learning-based methods.
- On the cross-area test set of the VIGOR dataset, our method significantly improves the localization accuracy of three state-of-the-art localization models in the target area, demonstrating its generalizability and effectiveness.

2 Related Work

2.1 Cross-View Localization

Cross-view image retrieval involves identifying the corresponding aerial image from a reference aerial image library for a given ground query image, using the center of the retrieved image as the estimated location for the ground query image. [5, 22] used handcrafted feature extractors to extract feature representations for both views and evaluated the similarity between the query image and the reference aerial image using the Euclidean or cosine distances between the features [54]. Previous deep learning works [4, 15, 23, 33, 41, 47, 49, 56, 62, 66] employed Siamese networks to extract features for both views separately, but they overlooked the significant domain gap between them. To bridge this gap, some works applied geometric transformations to the input images [6, 21, 29, 37, 39, 40] or used GANs to synthesize one view from another [32, 33, 47]. More recently, to improve cross-area generalization and robustness, researchers have explored techniques such as learning disentangled geometric layouts [64, 65], hard negative sampling strategies [8], and enforcing feature invariance to ground view variations [30]. While these retrieval-based methods are effective

at coarse matching, they fundamentally assume the ground camera is located at the center of the retrieved aerial image, which is often not the case and limits their precision for fine-grained applications.

Cross-view fine-grained localization builds upon image retrieval by further estimating the pixel-level location of the ground camera in the aerial image. [67] introduced the VIGOR dataset for this task, formulating it as a regression problem, where global feature descriptors from both views are used to regress the query camera’s location. [60] treats the task as a multi-class classification problem, dividing the aerial feature map into grid volumes and estimating the camera position based on the similarity between the global ground features and the local features of the aerial volumes. [17] and [57] incorporated geometric information by dividing the ground image features into equally sized sub-volumes along the horizontal direction, with each sub-volume corresponding to a different camera orientation. For each candidate coordinate, the similarity between each directional sub-volume of the ground feature and the corresponding local aerial feature is computed, and the candidate coordinate with the highest similarity is selected as the estimated camera position. Inspired by such geometric approaches, our instance matching step incorporates instance-level geometric and semantic information as cues for localization. [35] transforms the aerial features into the ground view and iteratively estimates the camera pose using Levenberg-Marquardt algorithm. [13, 38, 43], on the other hand, transform the ground features into the aerial view and compare them with aerial features at different locations to estimate the camera position. Our view alignment step employs two sets of geometric transformations to convert between the ground and aerial views. Although these learning-based fine-grained localization methods perform well in the source area, their performance degrades significantly when directly generalized to target areas where GT is unavailable.

2.2 Unsupervised Domain Adaptation

Domain adaptation is a special form of transfer learning, widely applied in visual tasks [51], aimed at addressing the data distribution discrepancy between the source and target domains, and studying how to effectively transfer the knowledge learned from the source domain to the target domain. Unsupervised domain adaptation refers to transferring the model from the source domain to the target domain when labeled data is unavailable in the target domain [10].

Feature alignment methods [7, 27, 44, 48] reduce the distribution discrepancy between the source and target domains in the feature space through adversarial training and domain discriminators, bringing the features of both domains closer. These pre-adaptation methods have shown good performance in single-image visual tasks such as object detection [68], image classification [28], and face recognition [1]. However, for cross-view fine-grained localization tasks, existing methods usually calculate the similarity between local features of the aerial image at different positions and the local features of the ground image, generate uncertainty heatmaps, and improve the similarity of ground-aerial features at GT locations through contrastive learning. Therefore, minimizing the global feature mean discrepancy between the target and source domains through adversarial training is not suitable for cross-view localization tasks.

Pseudo-labeling methods use the output of a model trained on the source domain as pseudo-GT for the target domain. However, the model’s output in the target domain often exhibits greater uncertainty [3, 42, 50]. [61] found that while entropy minimization can enhance the confidence of the model’s output, it can actually exacerbate localization errors. They use a teacher model pre-trained in the source domain to generate initial pseudo-GT for an auxiliary student model. Subsequently, only the predictions on which both the teacher and the auxiliary student agree are used as high-confidence pseudo-GT to train the final student model. Since all three models (teacher, auxiliary student, and student) share the same network architecture, the teacher and auxiliary student models are likely to produce identical erroneous predictions, which leads to the student model being influenced by erroneous pseudo-labels. The approach proposed in this paper generates high-quality pseudo-GT using multiple teacher models and the CVMA module.

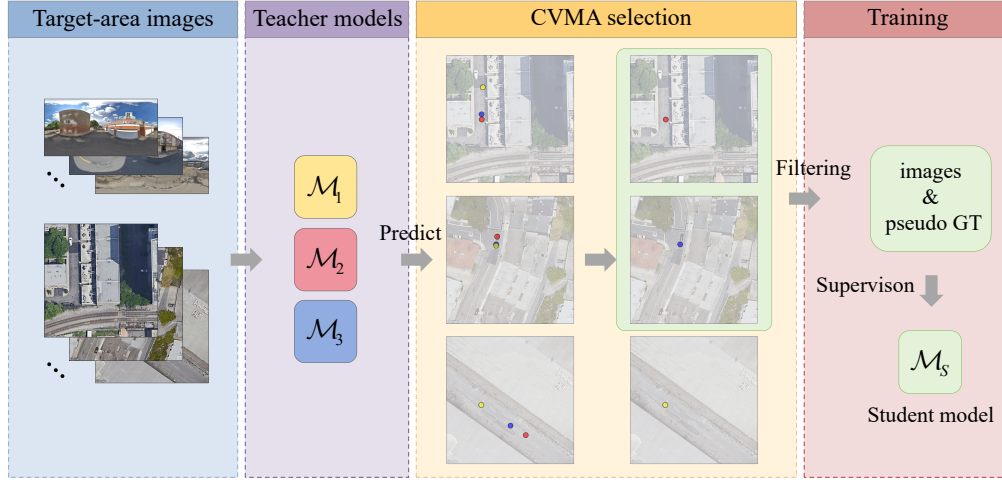


Fig. 2. Overview of the proposed Multi-Teacher Knowledge Distillation Approach. First, multiple teacher models pre-trained on the source domain are used to predict the camera coordinates for the target domain images. Next, the predictions from different teachers are evaluated by CVMA, and these predictions are selected and filtered based on their CVMA scores. Finally, the filtered target domain images and their corresponding pseudo-GT are used to train the student model.

3 Methodology

This section provides an overview of the weakly supervised learning approach based on multi-teacher knowledge distillation, and details its CVMA module.

3.1 Approach Overview

Given a pair of matching ground image G and aerial image A , the cross-view fine-grained camera localization task aims to estimate the camera’s coordinates $\hat{y}$ in the aerial image. Directly applying models trained in a source area to a new target area (cross-area setting) often leads to a performance drop. Our goal is to adapt the fine-grained localization model to the target area when the true camera location y in the target area is unknown. Following the standard fine-grained localization setting, we adopt the common assumption of known camera orientation [24, 41, 60, 61, 63, 67], as vehicles and robots can easily obtain orientation using magnetometers and IMUs. Thus, our task is strictly focused on estimating the camera’s 2D position, not its orientation.

Compared to directly using the prediction from a single teacher model as pseudo-GT (knowledge self-distillation), multi-teacher knowledge distillation provides a richer and more comprehensive source of knowledge [26], which is expected to improve the reliability of pseudo-GT. As shown in Figure 2, we use a set of teacher models pre-trained in the source area $\mathcal{M}_j$ (where $j = 1, 2, \dots, J$) to predict the camera coordinates $\hat{y}_j$ for each ground-aerial image pair $x_i = (G_i, A_i)$ (where $i = 1, 2, \dots, K$) in the target area. We use the proposed CVMA module to evaluate the quality of the camera coordinates predicted by different teacher models, and the best prediction $\hat{y}_i^*$ is selected.

$$\hat{y}_i^* = \arg \max_{\hat{y}_j} \text{CVMA}(\hat{y}_j). \quad (1)$$

The work of [61] found that fewer but more reliable samples from the target area are more beneficial for the student model’s performance. To filter out potential outlier samples caused by model prediction errors and noisy data, we sort

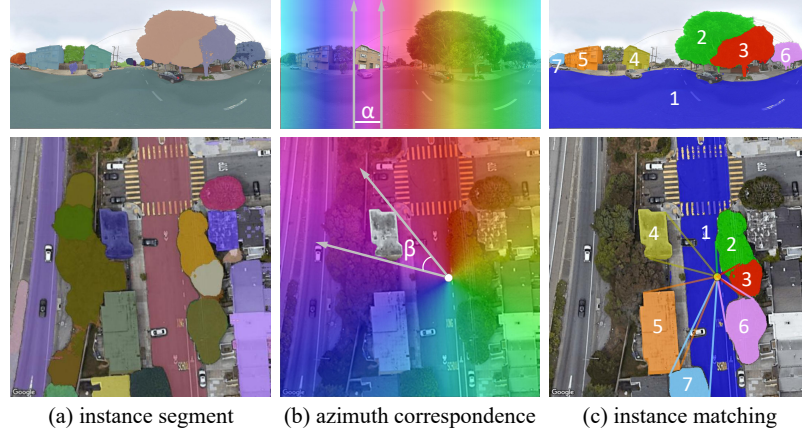


Fig. 3. Matching of static instances in the segmented ground-aerial image pairs (a), based on the semantics and azimuth IoU of the ground-aerial instances (b). The matching results for the candidate coordinates are shown in (c), with matching instance pairs represented in the same color.

the sample set from the target area $\{(x_1, y_1^*), (x_2, y_2^*), \dots, (x_K, y_K^*)\}$ based on their CVMA scores and retain only the top-k% of samples.

Different models vary in their architectures and loss function designs, thus requiring distinct hierarchical supervision signals. This makes it impractical to directly use the teacher models' output heatmaps as hierarchical pseudo-GT for the student. We therefore generate the full set of pseudo-GT using the optimal coordinate $\hat{y}^*$, following the student model's original manner.

Essentially, our proposed method functions as a pseudo-GT generation mechanism that does not alter the model's architecture or training paradigm. As a result, the training and inference speeds of the student model remain unaffected, and no additional overhead is incurred at runtime.

3.2 CVMA

The instance matching score $S_{\text{match},j}$ and the view alignment score $S_{\text{align},j}$ for the candidate predicted camera coordinates $\hat{y}_j$ are calculated, and the normalized scores are summed to obtain the CVMA score.

$$\text{CVMA}(\hat{y}_j) = \text{norm}(S_{\text{match},j}) + \text{norm}(S_{\text{align},j}). \quad (2)$$

3.2.1 Instance Matching. Ground-aerial image pairs represent different perspectives of the same scene, where the ground-aerial instances of the same static object have corresponding azimuth relationships with the camera and are semantically consistent. Based on the azimuth and semantics of the instances, we perform ground-aerial instance matching at the candidate coordinate $\hat{y}$, and calculate the instance matching score using the azimuth intersection over union (IoU) for each matching pair. The vertical lines in the panoramic ground image correspond to rays starting from the camera location in the aerial image [37, 40]. Therefore, the closer the candidate camera coordinate $\hat{y}_j$ is to the true camera coordinate y , the higher the instance matching score. When there are enough static reference objects in the scene, the instance matching score effectively reflects the proximity between each candidate coordinate $\hat{y}_j$ and the true camera coordinate y .

The matching process is illustrated in Figure 3. First, based on the text prompt, the ground image G and aerial image A are instance-segmented to obtain static instance sets for the two views, $\mathcal{I}^G = \{I_1^G, I_2^G, \dots, I_M^G\}$ and $\mathcal{I}^A = \{I_1^A, I_2^A, \dots, I_N^A\}$. Next, the edge contours of all instances are extracted to determine the tangents. The two vertical tangent lines to the ground instance I_m^G determine the relative azimuth α_m^G between the ground instance and the ground camera. The two rays, starting from the candidate coordinate $\hat{y}_j$ and tangent to the aerial instance I_n^A , determine the relative azimuth $\beta_{n,j}^A$ between them. Then, for each ground instance I_m^G , the matching aerial instance is determined based on semantic consistency and an azimuth IoU above a threshold t_{IoU} . If multiple matching aerial instances meet the conditions, the one with the highest IoU is selected as the corresponding aerial instance I_n^A , and the index pair (m, n) is recorded in the matching set M_j . Finally, the matching score $S_{\text{match},j}$ for the candidate coordinate $\hat{y}_j$ is computed.

$$\text{IoU}(\alpha, \beta) = \frac{\alpha \cap \beta}{\alpha \cup \beta}, \quad (3)$$

$$S_{\text{match},j} = \sum_{(m,n) \in M_j} \frac{\text{IoU}(\alpha_m^g, \beta_{n,j}^a) - t_{\text{IoU}}}{1 - t_{\text{IoU}}}. \quad (4)$$

3.2.2 View Alignment. Geometric projection transformations enable the mutual conversion between the ground view and the aerial view. We generate a set of ground-aerial views separately from the ground image G and the aerial image A centered at each candidate coordinate $\hat{y}_j$, as shown in Figure 4. The true camera coordinates y in the ground image G and the aerial image A have a homography relationship on the ground plane. Therefore, the closer the candidate coordinate $\hat{y}_j$ is to the true camera coordinate y , the higher the visual similarity between the two sets of ground-aerial views. When the ground plane in the scene contains visual information such as road edges and landmarks, view alignment effectively reflects the proximity between each candidate coordinate $\hat{y}$ and the true camera coordinate y .

Specifically, for a ground image G of size $H_G \times W_G$, spherical transformation T [53] and cropping-resizing operations are applied to generate the transformed ground image G^T and the ground reference image G^R . The mapping relationship between the transformed image coordinates (u^T, v^T) and the ground image coordinates (u^G, v^G) is given by Equation 5. On the other hand, using the candidate coordinates $\hat{y}_j = (\hat{u}_j, \hat{v}_j)$ as the geometric center, the aerial image A of size $L_A \times L_A$ undergoes a polar transformation T' [40] and cropping-resizing operations to generate the transformed aerial image $A_j^{T'}$ and the aerial reference image A_j^R . The mapping relationship between the transformed image coordinates $(u^{T'}, v^{T'})$ and the aerial image coordinates (u^A, v^A) is described in Equation 6, where z is a scaling factor representing the ground sampling distance. The purpose of the cropping-resizing operations is to ensure that the field of view and size of the reference image are consistent with the transformed image. The field of view (FoV) for the geometric transformation is set to 80° . Finally, two pairs of images $P_j = \{(G^T, A_j^R), (G^R, A_j^{T'})\}$, each with size $L \times L$, are used to evaluate the alignment effect of each candidate coordinate $\hat{y}_j$.

$$\begin{cases} p = L/2 - u \\ q = L/2 - v \\ f = L/2 \tan(\text{FoV}) \\ u^G = [1 - \arctan 2(p, q) / \pi] W_G / 2 \\ v^G = \left[0.5 - \arctan 2 \left(-f, \sqrt{p^2 + q^2} \right) / \pi \right] H_G, \end{cases} \quad (5)$$

$$\begin{cases} u^A = \hat{u}_j + zL_A \tan(\pi v' / L) \cos(2\pi u / L) \\ v^A = \hat{v}_j - zL_A \tan(\pi v' / L) \sin(2\pi u / L) \end{cases} \quad (6)$$

To assess the structural and textural similarity of an image pair (g, a) , we employ two complementary metrics. Specifically, we use SSIM to measure structural similarity. Additionally, we design an improved metric based on ORB feature matching. Unlike standard matching which counts correspondence, our metric quantifies alignment quality by calculating an ORB score from the set of L2 distances, $D_{(g,a)} = \{d_1, d_2, \dots\}$, between matched feature points, with a distance-weighted mechanism as defined in the following equation:

$$\text{ORB}(g, a) = \sum_{\substack{d \in D_{(g,a)} \\ d < t_d}} \exp(-\gamma \cdot d), \quad (7)$$

where the parameter t_d is the feature point matching distance threshold, and γ is the distance decay factor. The alignment score for the candidate coordinates $\hat{y}$ is obtained by summing the scores of the two metrics from the two sets of image pairs, as shown in Equation 8:

$$S_{\text{align},j} = \sum_{(g,a) \in P_j} [\text{SSIM}(g, a) + \text{ORB}(g, a)]. \quad (8)$$

We select SSIM and ORB as similarity metrics because they rely on low-level visual primitives (structure and corners) and are independent of any specific deep learning model. This independence is crucial for generating unbiased pseudo-labels, avoiding the potential domain bias introduced by pre-trained deep feature extractors. We acknowledge that traditional metrics can be sensitive to illumination changes and seasonal variations. This limitation, however, is effectively mitigated by our Instance Matching module, which compensates for visual inconsistencies under drastic appearance changes. While advanced deep features could offer stronger descriptors, utilizing training-free traditional metrics aligns better with the core philosophy of unsupervised domain adaptation.

In summary, the CVMA module provides a comprehensive and robust quality assessment for candidate coordinates through two complementary scoring dimensions. Instance Matching leverages macroscopic geometric and semantic consistency, excelling in scenes with prominent static landmarks. View Alignment, in contrast, focuses on microscopic visual textures and local features, making it particularly sensitive to scenes rich in ground-plane information, such as road markings. Crucially, these two modules are synergistic: when one module is less effective due to scene characteristics (e.g., sparse landmarks or view occlusion), it tends to produce a flat score distribution with low variance (low certainty); in contrast, if the other module successfully captures valid cues, it yields a peaked distribution with high variance (high certainty), thereby dominating the final fused score. This mechanism ensures that CVMA dynamically relies on the most reliable cues available in the current scene to generate high-quality pseudo-labels.

4 Experiment

We first introduce the dataset and evaluation metrics used in this study, followed by the state-of-the-art fine-grained localization models employed in the experiments. We then provide detailed implementation details. Finally, we present the cross-region testing results of our method and provide the ablation study results.

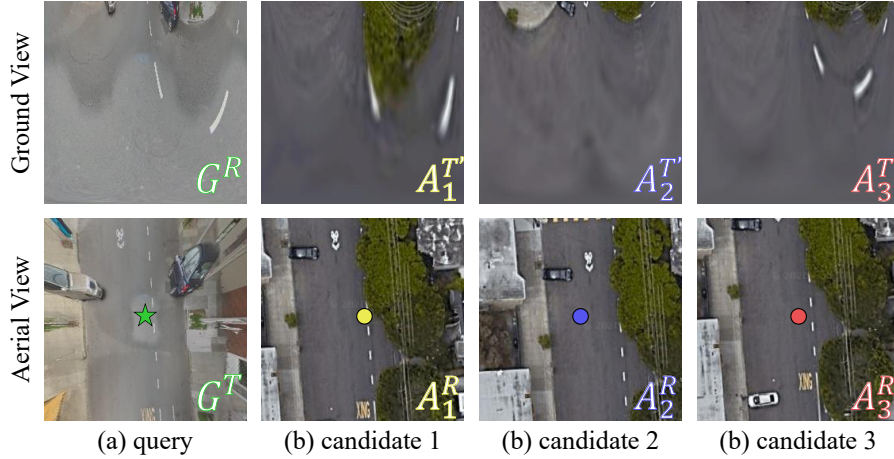


Fig. 4. Ground-aerial views generated from the queried ground panoramic image (a) and from the aerial image centered at three candidate points (b), (c), (d).

4.1 Dataset and Evaluation Metrics

Compared to ground images with a limited FoV, panoramic images contain more complete scene information and richer localization cues, which makes them better suited for our proposed learning-free CVMA module to generate high-quality pseudo-labels. Therefore, we chose the VIGOR dataset [67] for our experiments, rather than datasets with limited FoV such as KITTI [14]. The VIGOR dataset consists of ground panoramic images and aerial images collected from four cities in the United States: New York, Seattle, San Francisco, and Chicago. The cross-area setting of VIGOR uses all data from New York and Seattle for training, and data from San Francisco and Chicago for testing. Each ground panoramic image in VIGOR is spatially covered by four aerial images, including one positive aerial image and three semi-positive aerial images. The ground camera is located within the central quarter of the positive aerial image. The panoramic images and aerial images are orientation-aligned, meaning that the central vertical line of the panoramic image corresponds to the north direction in the aerial image. In our experiments, we use the positive aerial images and assume a known orientation. This assumption is practical for real-world scenarios, as outdoor robots or vehicles can readily obtain directional priors and coarse localization from magnetometers and GNSS.

Following the setting in [61], we split the cross-area test set into 70%, 10%, and 20% for the training set (without ground-truth camera locations), validation set, and test set, respectively. We train the student model using the ground-aerial image pairs from the training set and the pseudo-GT generated by us. Ablation studies are conducted on the validation set to determine the early stopping epoch. The model’s performance in direct generalization and adaptation through the weakly supervised learning approach is compared on the test set. We utilize the GT camera locations corrected by [17] and the city-specific ground resolution they provide. The evaluation metrics include the mean and median localization errors (in meters) between the predictions and ground-truth across all samples in the test set.

4.2 State-of-the-Art Baseline Models

We employ three state-of-the-art (SOTA) fine-grained localization models in our experiments: CCVPE [57], GGCVT [38], and HC-Net [53]. These models play a dual role. First, they act collectively as a teacher ensemble to generate

Table 1. Evaluation on the VIGOR test set.

model	DG		SKD		MTKD (Ours)	
	Mean (m)	Median (m)	Mean (m)	Median (m)	Mean (m)	Median (m)
CCVPE[57]	5.12	1.72	4.91 (↓ 4.10%)	1.63 (↓ 5.23%)	3.34 (↓ 34.77%)	1.28 (↓ 25.28%)
GGCVT[38]	5.30	1.43	5.12 (↓ 3.40%)	1.45 (↑ 1.40%)	3.64 (↓ 31.32%)	1.19 (↓ 16.78%)
HC-Net[53]	3.38	1.60	3.04 (↓ 10.06%)	1.48 (↓ 7.50%)	2.79 (↓ 17.46%)	1.17 (↓ 26.88%)

candidate predictions for the target area. Second, after our CVMA module selects the optimal prediction to create pseudo-GT, each model individually acts as a student, which is then trained on the filtered pseudo-GT samples to adapt to the target area.

4.3 Implementation Details

All experiments were conducted on a workstation equipped with an Intel Xeon W-2223 CPU and a single NVIDIA RTX 3090 GPU. In the instance matching step, we employ text-prompted Grounding DINO [25] to generate bounding boxes, which serve as prompts for SAM 2 [16] to perform fast instance segmentation. Both the box threshold and text threshold are set to 0.25. The azimuth IoU threshold t_{IoU} is set to 0.5; preliminary experiments on the validation set indicate that varying this threshold within [0.4, 0.6] has negligible impact on pseudo-label quality, demonstrating the robustness of our method. In the view alignment step, the field of view (FoV) for geometric transformation is set to 80°, a value chosen to maximize the effective visual content overlap between the transformed cross-view images. The generated image pairs are resized to 256×256 . The window size for SSIM is set to 31×31 , and for the ORB alignment metric, the distance threshold t_d is 30 with a decay factor $\gamma = 0.01$. The sample retention percentages k for CCVPE, GGCVT, and HC-Net are set to 90%, 80%, and 60%, respectively. All student models are initialized with their respective weights pre-trained on the source domain. This follows the standard practice for domain adaptation via fine-tuning.

4.4 Results

On the split test set, we evaluated the performance of three SOTA baseline models under direct generalization (DG), self-knowledge distillation (SKD), and multi-teacher knowledge distillation (MTKD). SKD refers to using each model’s own predictions in the target area as pseudo-GT to supervise model training, and we introduce this method as an additional comparison reference. MTKD is the weakly supervised learning approach proposed in this paper.

The experimental results demonstrate that our multi-teacher knowledge distillation method significantly outperforms the self-knowledge distillation method in improving cross-domain performance, as shown in Table 1. Due to the limited quality of pseudo-GT in the SKD method, which is constrained by the model’s own direct generalization capability, the reductions in mean and median error for the three baseline models are minimal, and the median error of GGCVT even increases by 1.4%. In contrast, our MTKD method substantially reduces the localization error of all three models in the target region. Specifically, for CCVPE, MTKD reduces the mean and median errors by 34.77% and 25.28%, respectively; for GGCVT, the reductions are 31.32% and 16.78%, respectively; and for HC-Net, the reductions are 17.46% and 26.88%, respectively.

It is worth noting that our implementation of Self-Knowledge Distillation (SKD) as a baseline is conceptually similar to the work of [61]. Since their source code and specific data splits were not publicly available, a direct numerical comparison on the exact same test set was not feasible. Therefore, while we followed their splitting ratio to construct

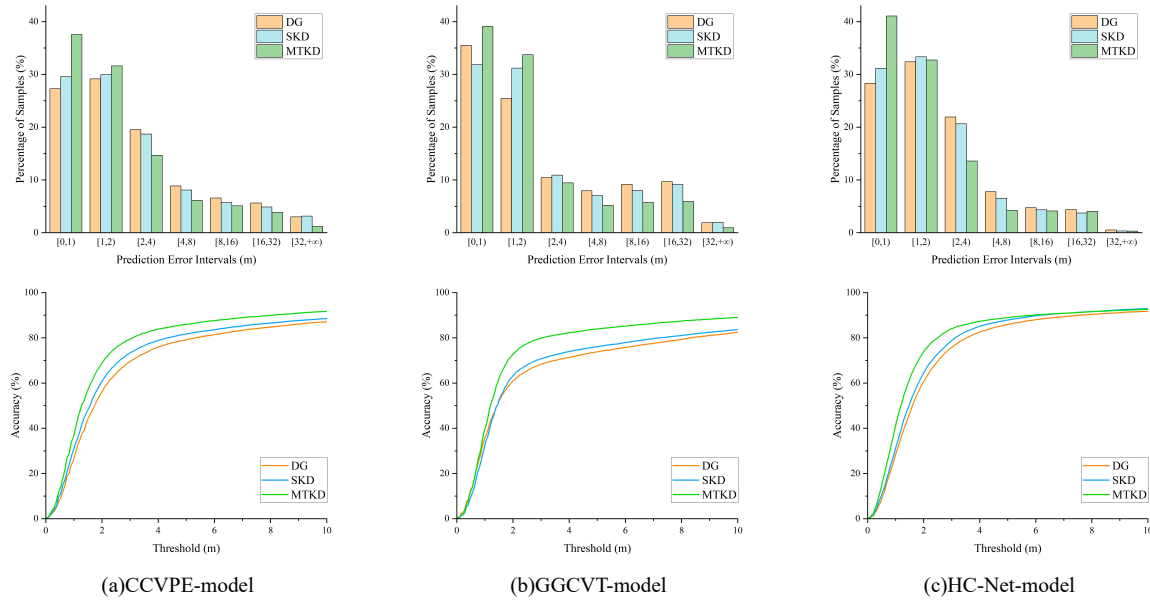


Fig. 5. Comparison of localization error distributions on the cross-area test set. Top row: Error Interval Distribution for three models (HC-Net, CCVPE, GGCVT), showing the percentage of samples within each error range. Bottom row: Cumulative Error Distribution curves, illustrating the localization accuracy at varying distance thresholds.

our test set, we conduct an indirect comparison by evaluating the relative performance improvement. [61] reported that their method reduced the mean and median errors of GGCVT by 16% (from 5.19m to 4.34m) and 5% (from 1.39m to 1.32m), respectively. In contrast, under our setting, our MTKD method achieved a more substantial reduction of 31.32% and 16.78% for the same model. Similarly, their reported improvements for CCVPE were 12% and 11%, whereas our method achieved 34.77% and 25.28%. This comparison, despite the differences in test sets, strongly suggests that our multi-teacher distillation framework offers a more significant advantage in enhancing model generalization than a single-teacher self-distillation approach.

We analyze the error interval distribution and cumulative error distribution, as illustrated in Figure 5. To facilitate analysis, we employ a geometric progression-based binning criterion for defining error intervals. Specifically, the initial bin is set to $[0,1)$, and subsequent bin upper bounds are incremented multiplicatively: $[1,2)$, $[2,4)$, $[4,8)$, $[8,16)$, $[16,32)$, with an open-ended interval $[32,+\infty)$ to cover extreme error values. Compared to DG, our MTKD method significantly increases the proportion of samples in low-error intervals while reducing the proportion in high-error intervals. When the error threshold is set to 1m, the accuracy of CCVPE under direct generalization is 27.27%, which improves to 37.54% after MTKD weak supervision. For GGCVT, MTKD improves accuracy from 35.49% (under DG) to 39.09%, whereas SKD even decreases accuracy to 31.81%. For HC-Net, the accuracy under DG is 28.32%, while MTKD boosts it to 41.05%.

We compared the HC-Net adapted through MTKD with two other weakly supervised models [36] and [52]. We consider their reported results under two distinct settings: one that assumes coarse GT is available, and another that, like ours, assumes no GT data is available at all. Additionally, we also present the performance of state-of-the-art localization models in cross-area direct generalization for comparison. As shown in Table 2, our MTKD-adapted HC-Net achieves the best performance among all compared methods on the cross-area test set.

Table 2. Comparison with other Localization works

	Method	Cross-Area Test	
		Mean (m)	Median (m)
Supervised	MCC[60]	13.06	6.31
	FF-Net[63]	7.28	2.83
	SliceMatch[17]	5.53	2.55
	CCVPE[57]	4.97	1.68
	GGCVT[38]	5.16	1.40
	DenseFlow[43]	5.01	2.42
	C-BEV [12]	3.97	2.86
	HC-Net[53]	3.35	1.59
	C2F-CCPE[45]	3.94	1.68
Weakly	G2S-Weakly [36]	5.37	1.93
	BevSplat [52]	4.50	1.95
	G2S-Weakly* [36]	4.70	1.68
	BevSplat* [52]	3.80	1.70
	HC-Net-MTKD(Ours)	2.79	1.17

*Note: Methods marked with * assume the availability of rough GT for supervision.*

4.5 CVMA Localization

To validate the effectiveness of CVMA, we treat it as a learning-free cross-view fine-grained localization method to predict camera coordinates on the VIGOR cross-area test set. First, we perform grid sampling on the aerial image, compute the CVMA score for each sampled coordinate, and normalize the scores. Then, for coordinates with normalized scores greater than 0.9, we iteratively perform finer-grained sampling (with pixel spacing of 15, 5, 3, and 1) and compute the corresponding scores. Finally, the coordinate with the highest score is selected as the estimated camera position.

Our CVMA localization method achieves a mean error of 5.31m and a median error of 1.54m on the cross-area test set. Even without training, compared to SliceMatch [17] which also employs grid sampling but requires training, our method reduces the mean error by 4% and the median error by 39.6%. This performance gap stems from the differing sensitivity to positional variations. SliceMatch determines localization by assessing the co-occurrence of visual features within corresponding directional slices. However, for spatially adjacent candidate coordinates, the feature distribution within slices often remains largely invariant, making it difficult for the model to distinguish minute positional shifts. In contrast, CVMA leverages both geometric projection and instance azimuth constraints: even minor coordinate changes induce noticeable geometric distortions in the projected images and alterations in the relative azimuth IoU of matched instances. This dual sensitivity allows CVMA to capture subtle positional discrepancies more precisely, enabling finer-grained localization.

To further investigate the effectiveness of CVMA across different scenes, we visualized the score heatmaps of sampled coordinates and the matching results at the final predicted positions (Figure 6). Overall, CVMA performs exceptionally well in environments with abundant static reference instances or clear ground-plane textures, demonstrating significant complementarity between its instance matching and view alignment modules. Specifically, as shown in Figure 6(a), when both abundant static instances and clear road structures are present, the two cues work synergistically to produce a highly certain heatmap peak. In cases where one cue is missing, the other module often provides critical compensation:

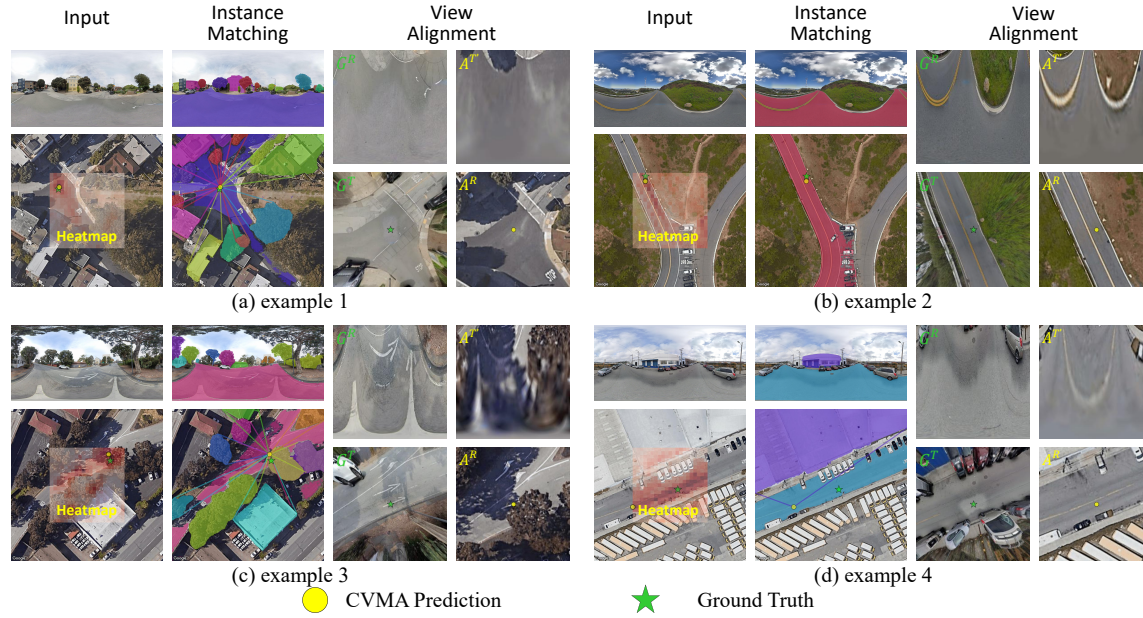


Fig. 6. Each example presents the following from left to right: (1) the CVMA score heatmap, where the red shading indicates the predicted probability, the yellow circle denotes the predicted camera position, and the green pentagram denotes the ground truth camera position; (2) the visualization of instance matching at the predicted camera position; (3) two pairs of images generated during view alignment at the predicted coordinates.

in the instance-scarce scene of Figure 6(b), CVMA relies on view alignment of road markings for localization; conversely, in Figure 6(c) where overhead occlusion interferes with view alignment, abundant instance matching ensures localization accuracy. However, CVMA still has limitations in extremely challenging sparse scenes. As shown in Figure 6(d), when instance segmentation fails due to ambiguous building edges and the environment lacks recognizable visual landmarks, the simultaneous scarcity of both cues leads to uncertainty in the heatmap, resulting in large localization errors. This explains why CVMA excels in median error (representing typical scenarios) but still lags behind learning-based SOTA methods in mean error, which is sensitive to extreme outliers.

4.6 Computational Efficiency Analysis

To comprehensively evaluate the practicality of the proposed method, we analyze the computational cost of the MTKD framework during the offline domain adaptation phase and the runtime efficiency of CVMA as a standalone locator. The detailed time consumption for each specific module is provided in Table 3.

Offline Adaptation Overhead. The generation of pseudo-GTs involves a one-time instance segmentation and multiple evaluations of teacher predictions. For a configuration with J teacher models, the total time required to process a single sample is approximately $T_{\text{total}} = T_{\text{seg}} + J \times T_{\text{eval}}$. In our experimental setup ($J = 3$), generating a single pseudo-GT takes about 0.78s. It is worth emphasizing that this constitutes a strictly offline data preprocessing step. Once the pseudo-GTs are generated, the fine-tuning and online inference processes of the student model in the target domain are entirely unaffected. Since the network architecture of the student model remains unchanged, its inference speed remains identical to that of the original SOTA model, achieving zero extra online computational overhead.

Table 3. Detailed computational time breakdown for each module.

Module	Time (ms)
Instance Segmentation (T_{seg})	652.67
<i>CVMA Evaluation Components:</i>	
Instance Matching	10.55
Transform & Crop	7.47
ORB Calculation	7.62
SSIM Calculation	18.04
Total per Candidate (T_{eval})	43.68

Table 4. Ablation study for CVMA

settings	Cross-Area Validation Set	
	Mean (m)	Median (m)
Match+Align	5.39	1.48
Match+Align-ORB	5.57	1.57
Match+Align-SSIM	6.73	1.75
Match+Align-G	5.86	1.48
Match+Align-A	5.91	1.99
Match	7.24	3.23
Align	9.78	6.68

CVMA Standalone Localization Efficiency. When CVMA is employed as a learning-free locator, the time consumption depends on the total number of sampled points N_{pts} . Adopting the hierarchical sampling strategy described in Section 4.5, the average number of sampled points N_{pts} is approximately 145. The total time for a single localization query is $T_{\text{loc}} = T_{\text{seg}} + N_{\text{pts}} \times T_{\text{eval}}$, which amounts to approximately 7.0s. For comparison, the single-frame inference times of learning-based SOTA models, such as CCVPE and GGCVT, are 0.22s and 0.17s, respectively. Although CVMA is significantly slower than end-to-end models and thus less suitable for real-time navigation, its core value lies in its learning-free nature. This makes it uniquely suitable for performing initial localization in unseen areas with zero training data (cold start) or automatically generating pseudo-labels for new domains to train efficient student models.

4.7 Ablation Study

We conduct comprehensive ablation studies to verify the contribution of each module component and analyze the impact of key hyperparameters.

Effectiveness of CVMA Components. We first evaluate the impact of each component within the CVMA module on coordinate quality assessment. Specifically, we define the following experimental settings: **Match+Align** (proposed): the default configuration that includes both instance matching and view alignment, where view alignment evaluates the visual similarity of two sets of image pairs $\{(G^T, A^r), (G^r, A_j^{T'})\}$ using SSIM and ORB. **Match+Align-ORB**: removes the ORB metric from the view alignment process and uses only SSIM as the visual similarity measure. **Match+Align-SSIM**: removes SSIM and relies solely on ORB to evaluate visual similarity. **Match+Align-G**: performs view alignment only on the image pair $(G^r, A_j^{T'})$, while **Match+Align-A**: performs view alignment only on (G^T, A^r) . **Match**: uses only instance matching to evaluate coordinate quality, whereas **Align**: uses only view alignment for coordinate evaluation.

Table 5. Ablation study on the impact of teacher diversity and quantity. We evaluate different teacher combinations for each student model. ✓ denotes the inclusion of the teacher model in the ensemble.

Student Model (Architecture)	Teacher Ensemble			Target Domain Error	
	HC-Net	CCVPE	GGCVT	Mean (m)	Median (m)
HC-Net[53]	✓			3.04	1.48
	✓	✓		3.07	1.43
	✓		✓	3.08	1.45
	✓	✓	✓	2.79	1.17
CCVPE[57]		✓		4.91	1.63
	✓	✓		3.58	1.34
		✓	✓	3.40	1.36
	✓	✓	✓	3.34	1.28
GGCVT[38]			✓	5.12	1.45
	✓		✓	3.74	1.20
		✓	✓	3.81	1.20
	✓	✓	✓	3.64	1.19

For each of the above settings, we evaluate the sampled coordinate set in the aerial image and select the coordinate with the highest score as the estimated camera position. The distance errors under different settings are summarized in Table 4.

Impact of Teacher Quantity and Diversity. To validate the effectiveness of the multi-teacher strategy, we evaluated different combinations of teacher quantity and types on the test set, as summarized in Table 5. The results indicate that the performance of all student models shows a consistent upward trend as the number of high-quality teachers increases. The essence of this improvement stems from the enhancement of pseudo-label quality. Introducing multiple heterogeneous teachers significantly enriches the diversity of the candidate prediction pool, thereby markedly increasing the probability that "at least one candidate prediction is close to the true location." However, this does not imply that the number of teachers can be increased indefinitely; the quality of teachers is more critical than their quantity. Introducing low-quality teachers may inject noise into the candidate pool, increasing the risk of CVMA selecting incorrect predictions. Conversely, excessively stacking high-quality teachers with highly consistent predictions would yield diminishing returns due to a lack of diversity, while incurring unnecessary computational costs. In this work, ensembling three SOTA models represents an optimal trade-off between prediction diversity and computational efficiency.

Sensitivity to Sample Retention Rate k . Finally, consistent with [61], we further conducted an ablation study on the sample retention percentage k . As shown in Figure 7, we explored the optimal k for each model on the validation set. The experiments show that CCVPE, GGCVT, and HC-Net achieve their best performance at $k = 90\%$, 80% , and 60% , respectively. This reveals a trade-off between the quantity and quality of training samples: too few samples may lead to overfitting, while including too many outliers can have a negative impact on model performance. We hypothesize that the variation in optimal k values is related to the architectural characteristics of each model. Specifically, CCVPE achieves optimal performance at a high retention rate ($k = 90\%$), which may be attributed to its probability heatmap-based output paradigm that likely allows the model to tolerate uncertainty in pseudo-labels. In contrast, HC-Net performs best with a strict filtering strategy ($k = 60\%$). A plausible explanation is that its reliance on homography matrix regression

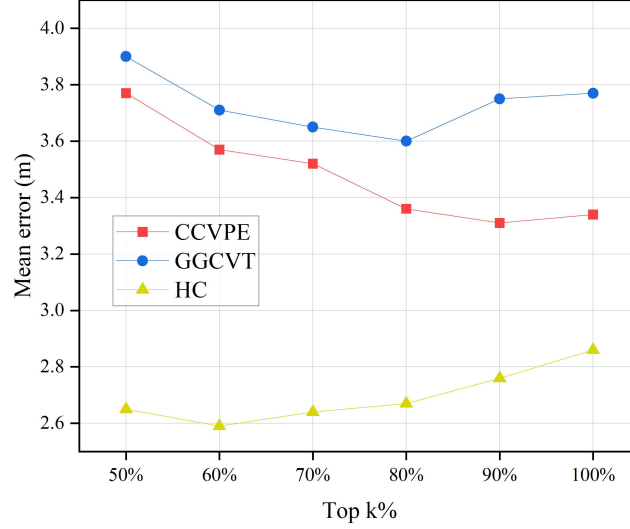


Fig. 7. Effect of sample retention percentage k . 100% means no sample filtering.

imposes strong geometric constraints, making the model highly sensitive to minor coordinate errors in pseudo-GTs. GGCVT ($k = 80\%$) falls in between, as its Transformer-based decoupled pose optimization seems to offer moderate noise resilience.

5 Limitation and Future Work

First, the current framework is specifically designed for panoramic images; adapting it to handle the more common and challenging scenario of limited FoV images remains a key next step.

Second, the instance matching component is highly dependent on the performance of the upstream instance segmentation model. As illustrated in the failure case in Figure 6(d), our method may struggle in scenes with ambiguous object boundaries (e.g., due to poor lighting or occlusion), where the segmentation model fails to produce reliable masks. This highlights a key vulnerability. However, this also presents a clear path for improvement: as foundation models for vision continue to advance, integrating more powerful and robust segmentation techniques could significantly mitigate this category of failures. For developers, this implies that the performance of our system in real-world applications is tightly coupled with the quality of the chosen segmentation backbone.

Lastly, the reliance on traditional computer vision metrics for view alignment could be superseded by exploring more powerful, learning-based feature representations specifically designed to better capture cross-view visual similarity.

6 Conclusion

In this paper, we proposed a novel weakly-supervised framework to address the domain adaptation challenge in cross-view localization. By leveraging multi-teacher knowledge distillation and our designed learning-free CVMA module, our approach effectively generates high-quality pseudo-GTs from diverse teacher predictions through a comprehensive evaluation of geometric, semantic, and visual cues. This significantly enhances the generalization performance of mainstream localization models in target areas without ground truth. Experiments demonstrate the effectiveness of the

framework and validate the potential of CVMA as a standalone learning-free localization method. Currently designed for panoramic images, a critical future direction is extending this method to limited FoV images with incomplete information, which is essential for large-scale real-world deployment.

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